

Complete real Kaehler submanifolds

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Introduction

Throughout this presentation we denote by

$$f: M^{2n} \rightarrow \mathbb{R}^{2n+p}$$

a **real Kähler submanifold**. This means that (M^{2n}, J) is a connected Kähler manifold of complex dimension $n \geq 2$ isometrically immersed into Euclidean space with substantial codimension p even locally.

The positiveness of the index of relative nullity imposes strong restrictions on complete a submanifold M of Euclidean space. Recall that the *index of relative nullity* $\nu(x)$ of f at $x \in M^{2n}$ is the dimension of the relative nullity tangent subspace defined by

$$\Delta(x) = \mathcal{N}(\alpha)(x) = \{X \in T_x M : \alpha(X, Y) = 0 \text{ for all } Y \in T_x M\},$$

where $\alpha: TM \times TM \rightarrow N_f M$ stands for the second fundamental form of f .

Let $f : M^n \rightarrow \mathbb{R}^m$ an isometric immersion whose index of relative nullity is positive at any point.

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Theorem (Hartman, 1970)

Let M^n be a complete manifold with nonnegative Ricci curvature and let $f : M^n \rightarrow \mathbb{R}^m$ be an isometric immersion with minimal index of relative nullity $\nu_0 > 0$. Then f is a ν_0 -cylinder.

When M^{2n} is assumed to be Kaehler the presence of almost complex structure tends to have strong restrictions on the structure of the submanifold. However, so far most known results are in low codimension.

Theorem (Abe, 1974)

Let $f : M^{2n} \rightarrow \mathbb{R}^{2n+1}$ a complete real Kaehler hypersurface that either f is real analytic, or the scalar curvature of M^{2n} is everywhere non-negative or everywhere negative. Then f is a cylinder over a complete surface $g : N^2 \rightarrow \mathbb{R}^3$, that is, $M^{2n} = N^2 \times \mathbb{C}^{n-1}$ and $f = g \times id$ split, where $id : \mathbb{C}^{n-1} \cong \mathbb{R}^{2n-2}$ is the identity map.

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This result was extended by Florit and Zheng (cf. Florit and Zheng (2008)) without any assumption on scalar curvature of M^{2n} or real analyticity of f .

Theorem (Florit and Zheng, 2007)

Let $f : M^{2n} \rightarrow \mathbb{R}^{2n+1}$ a complete real Kaehler hypersurface. Then f is a cylinder over a complete surface.

Note that, as a consequence of Hartman's cylinder theorem since any isometric immersion $f : M^n \rightarrow \mathbb{R}^{n+p}$ with $\nu(x) \geq n - 1$ for all $x \in M^{2n}$ is flat, we have that f is a cylinder over a curve, when M^n is also complete.

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Theorem (Abe, 1972)

Let a holomorphic isometric immersion $f : M^{2n} \rightarrow \mathbb{C}^{n+p}$ of a complete Kaehler manifold in to the complex Euclidean space, with $\nu(x) \geq 2n - 2$ for all $x \in M^{2n}$, then f cylinder over a holomorphic curve.

Conjecture (Florit and Zheng, 2007)

Let $f : M^{2n} \rightarrow \mathbb{R}^{2n+p}$ be an isometric immersion of a complete Kahler manifold into Euclidean space, if $\nu(x) \geq 2n - 2$ for all $x \in M^{2n}$, then f cylinder over a surface.

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Theorem (Dajczer and Rodríguez, 1991)

If $f : M^{2n} \rightarrow \mathbb{R}^{2n+2}$ is a complete minimal real Kaehler submanifold, then one of the following possibilities holds:

- i) f is holomorphic.
- ii) f is completely complex ruled.
- iii) f is a $(2n - 4)$ -cylinder.

Moreover, if the scalar curvature of M^{2n} has subquadratic growth along geodesics, then f is either of type (i) or (iii).

Theorem (Florit and Zheng, 2008)

Let $f : M^{2n} \rightarrow \mathbb{R}^{2n+2}$ be a complete analytic real Kaehler submanifold, not everywhere minimal, then f is cylinder over complete complex surface.

Theorem

Let $f: M^{2n} \rightarrow \mathbb{R}^{2n+p}$, $n > p$, be a complete analytic real Kaehler submanifold. If f is not everywhere minimal and $\nu_s^c(x) < 2n - 2s$ at any $x \in M^{2n}$ for $1 \leq s \leq p - 2$, then f is a $(2n - 2p)$ -cylinder over a complete real Kaehler submanifold $g: N^{2p} \rightarrow \mathbb{R}^{2p+p}$.

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Note that by Theorem 1 in (cf. S. Chion and Dajczer (2023)) if $f: M^{2n} \rightarrow \mathbb{R}^{2n+p}$ is nowhere minimal the hypothesis on the complex s -nullities, $\nu_s^c(x) < 2n - 2s$ at any $x \in M^{2n}$ for $1 \leq s \leq p - 2$, implies that at some point $x \in M^{2n}$ or $\nu_{p-1}^c(x) \geq 2n - 2p + 2$ or $\nu_p^c(x) = \nu^c(x) \geq 2n - 2p$.

The *complex s-nullity* $\nu_s^c(x)$ of $f: M^{2n} \rightarrow \mathbb{R}^{2n+p}$ at $x \in M^{2n}$, $1 \leq s \leq p$, is

$$\nu_s^c(x) = \max_{U^s \subset T_{f(x)}^\perp M} \{\dim(\mathcal{N}(\alpha_{U^s}) \cap J\mathcal{N}(\alpha_{U^s}))\},$$

where U^s denotes an s -dimensional subspace of $T_{f(x)}^\perp M$ and $\alpha_{U^s} = \pi_{U^s} \circ \alpha$ being $\pi_{U^s}: T_{f(x)}^\perp M \rightarrow U^s$ the projection. Notice that $\nu_p^c(x) = \nu^c(x)$.

Let V^n and W^m denote (real) vector spaces of finite dimension n and m , respectively. Given a (maybe not symmetric) bilinear form $\beta: V^n \times V^n \rightarrow W^m$, we denote by

$$\mathcal{S}(\beta) = \text{span} \{ \beta(X, Y) : X, Y \in V^n \}$$

the subspace generated by β and its (right) kernel by

$$\mathcal{N}(\beta) = \{ Y \in V^n : \beta(X, Y) = 0 \text{ for all } X \in V^n \}.$$

A bilinear form $\beta: V^n \times V^n \rightarrow W^{p,p}$ is called *flat* if

$$\langle \beta(X, Y), \beta(Z, T) \rangle - \langle \beta(X, T), \beta(Z, Y) \rangle = 0$$

for all $X, Y, Z, T \in V^n$.

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for all $X, Y, Z, T \in V^n$. It is said that β is *null* when

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for all $X, Y, Z, T \in V^n$.

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$$\beta(X, Y) = (\alpha(X, Y), \alpha(X, JY)). \quad (1)$$

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$$\gamma(X, Y) = \alpha(X, Y) + \alpha(JX, JY) \quad \text{and} \quad \tilde{\gamma}(X, Y) = \alpha(X, Y) - \alpha(JX, JY).$$

Let $\beta^\gamma, \beta: V^{2n} \times V^{2n} \rightarrow W^{p,p}$ be the *Hermitian* and *symmetric* parts of β , the bilinear forms associated by (1) to $\gamma, \tilde{\gamma}$, respectively, that is,

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$$\beta^\gamma(X, Y) = (\gamma(X, Y), \gamma(X, JY)) \quad \text{and} \quad \beta^{\tilde{\gamma}}(X, Y) = (\tilde{\gamma}(X, Y), \tilde{\gamma}(X, JY)). \quad (2)$$

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Then

$$2\beta = \beta^\gamma + \beta^{\tilde{\gamma}}. \quad (3)$$

Definition

We say that $\beta^\gamma, \beta^{\tilde{\gamma}}$ satisfies the mixed flat conditions if holds:

$$\langle\langle \beta^\gamma(X, Y), \beta^{\tilde{\gamma}}(Z, T) \rangle\rangle = \langle\langle \beta^\gamma(X, T), \beta^{\tilde{\gamma}}(Z, Y) \rangle\rangle \quad (4)$$

for any $X, Y, Z, T \in V$.

Lemma

Let $f : M^{2n} \rightarrow \mathbb{R}^{2n+p}$ be a real Kaehler submanifold. Then, $\beta^\gamma, \beta^{\tilde{\gamma}}$ are flat bilinear forms. Moreover, satisfies the mixed flat conditions.

Corollary

Let $f : M^{2n} \rightarrow \mathbb{R}^{2n+p}$ be a complete real Kaehler submanifold, then for any $T \in D$, C_T is nilpotent,

$$\beta^{\tilde{\gamma}}(C_T X, Y) = \beta^{\tilde{\gamma}}(X, C_T Y) \in \mathcal{S}(\beta^{\gamma})^{\perp}, \text{ and } \beta^{\gamma}(C_T X, JY) = 0, \quad (5)$$

that is, the splitting tensor $C_T : D^{\perp} \rightarrow D^{\perp}$ satisfies

$$\alpha(C_T X, Y) = \alpha(X, C_T Y), \quad \forall X, Y \in TM, \quad (6)$$

$$C_T \circ J = J \circ C_T, \quad C_T^r = 0, \quad \text{Im } C_T \subset \Delta_J. \quad (7)$$

Corollary

Let $f : M^{2n} \rightarrow \mathbb{R}^{2n+p}$ be a complete real Kaehler submanifold, assume that one of the following holds:

$$(a) \Delta_J \subseteq \mathcal{N}(\beta^{\tilde{\gamma}}), \quad \text{or} \quad (b) \dim \mathcal{S}(\beta^{\gamma}) \geq 2p - 2.$$

Then, $C = 0$. That is, each connected component U_i of U_0 is isometric to a product

$$U_i = N^{2r} \times \mathbb{C}^{n-r}, \quad r \leq p,$$

and $f|_{U_i} = f' \times id$ split, where $f' : N^{2r} \rightarrow \mathbb{R}^{2r+p}$ is a real Kaehler submanifold, and $id : \mathbb{C}^{n-r} \cong \mathbb{R}^{2(n-r)}$ is the identity map.

Proof of Theorem 1

The key point is the following observation: if $\nu_s^c < 2n - 2s$ for $1 \leq s \leq p - 2$ then $\dim \mathcal{S}(\beta^\gamma) \geq 2p - 2$ and the result follows from Corollary above item (b).

Proof of Theorem 1

The key point is the following observation: if $\nu_s^c < 2n - 2s$ for $1 \leq s \leq p - 2$ then $\dim \mathcal{S}(\beta^\gamma) \geq 2p - 2$ and the result follows from Corollary above item (b). Let's show this, suppose the contrary, i.e., $\nu_s^c < 2n - 2s$ for $1 \leq s \leq p - 2$ and $\dim \mathcal{S}(\beta^\gamma) < 2p - 2$.

Observe that always holds

$$\nu_s^c \geq \dim \mathcal{N}(\beta^\gamma) \geq 2n - \dim \mathcal{S}(\beta^\gamma) \quad (8)$$

Thus we get that $\nu_{\xi}^c \geq 2n - 2s$, and this contradicts the assumptions on the complex s -nullities.

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In fact, since $\beta^\gamma = 0$ is equivalent to $\alpha(X, JY) = \alpha(Y, JX)$ for all $X, Y \in T_x M$ and this, in particular, is equivalent to f being minimal, we have that $\dim U^\gamma = s \geq 1$.

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Obrigado!!!

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